Benchmarking dependency measures for marker gene identification in multi-modal single-cell data

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1 Project

1.1 What is the scientific question you want to answer?

When identifying genes whose expression is informative of cell identity in single-cell data, practitioners typically use Pearson/Spearman correlation or one-vs-rest differential expression. However, the underlying statistical question, is gene g associated with cell type c, has several distinct answers depending on whether one accounts for (a) nonlinearity in the gene–identity relationship, and (b) confounding by other genes. We ask: do these two axes meaningfully change which genes are identified as cell-type markers in PBMC CITE-seq data, and if so, which axis matters more?

We frame this as a 2×2 comparison of dependency measures:

	Marginal	Conditional
Linear	Spearman correlation	Partial correlation
Nonlinear	Mutual information (KSG)	Integrated Gradients on MLP

The protein measurements from CITE-seq provide an independent validation channel: a true marker gene should be associated with the corresponding surface protein, not just the cell-type label. This lets us cross-validate the RNA-based rankings against direct molecular evidence rather than only against curated marker lists.

1.2 What inspired you to work on this project?

Standard scRNA-seq pipelines (Scanpy, Seurat) default to Wilcoxon or t-test for marker identification, methods that assume marginal, monotonic relationships. Recent work on lineage drivers (e.g. CellRank 2, [1]) uses correlation between gene expression and fate probabilities. Yet the explainable-AI literature has produced powerful tools, including Integrated Gradients, SHAP, and attention-based attribution, that could in principle answer the same question. Whether the added complexity is warranted on real single-cell data, and which property (nonlinearity vs. multivariate context) drives any gain, is to our knowledge not systematically benchmarked.

1.3 What types of data will you analyse?

CITE-seq data, that is, simultaneous measurement of (a) whole-transcriptome RNA via standard 10x scRNA-seq and (b) ∼228 surface protein markers via oligo-conjugated antibodies (ADT), both measured in the same single cells [2]. Two modalities, same study, trivial integration (early, vertical integration using shared cell barcodes). Cell-type labels are provided by the original publication's multimodal reference annotation at three levels of granularity (8, 30 and 57 categories.)

1.4 What computational/statistical methods do you plan to use to analyse the data?

Preprocessing and exploration follow the standard Scanpy/muon workflow: QC filtering, normalization (log1p for RNA, CLR for ADT), HVG selection, PCA, UMAP visualization, and cell-type label validation via protein markers. Because the data spans 8 donors, we assess donor batch effects on the UMAP and, where present, apply batch correction (Harmony on the RNA PCA embedding) before downstream analysis.

The core methods comparison (one-vs-rest, per cell type) consists of:

- Spearman correlation between gene expression and the binary cell-type indicator.
- Partial correlation from the pseudoinverse of the covariance matrix or a similar procedure.
- Mutual information via the Kraskov–Stögbauer–Grassberger kNN estimator [3] (`sklearn.feature_selection.mutual_info_classif`), on rank-transformed expression to mitigate the zero-dropout confound.
- Differentiable classifier sensitivity: train a small MLP to predict cell type from the full transcriptome or top PCs, then compute Integrated Gradients [4] attributions per gene per class via Captum.

Evaluation uses a protein-derived ground truth, avoiding curated marker databases (which are largely compiled from the differential-expression methods under test, and introduce gene-identifier matching problems). For each cell type c, we define a ground-truth driver set D_c from the ADT modality alone: proteins that are differentially elevated in c versus the rest (one-vs-rest contrast on CLR-normalized[1] counts), mapped to the gene(s) encoding them (e.g. CD3 protein to `CD3D`/`CD3E`/`CD3G`). This

[1] CLR: Centered Log-Ratio; a standard normalization applied on surface protein data (ADT data) in single-cell sequencing

protein-to-gene mapping is a small, one-time curation based on molecular fact rather than on prior differential-expression results, so it is independent of all four methods. Crucially, the four methods see RNA only; the protein modality is used solely to build D_c, keeping the evaluation a genuine cross-modal test: can RNA-derived scores recover the genes whose proteins independently mark the cell type?

Each method produces a total ranking of all genes for c, and we score recovery of D_c using the rank-based driver-recovery metric of [1] by calculating the relative area under the curve (AUC_{rel}), which allows us to express the recovery rate of the true marker genes as a single score between 0.5 and 1.0. This metric is parameter-free and is defined on exactly the output of all four methods. We report $\mathrm{AUC}_{\mathrm{rel}}$ per cell type together with $|D_c|$, and exclude cell types with $|D_c| \leq 1$.

This setup makes the research question concrete and decomposable. Because RNA and protein for the same gene are only moderately correlated, a gene may be a protein-confirmed marker yet correlate weakly with the cell-type label. The central question is whether such weak linear correlation also implies low mutual information, low partial correlation, and low classifier sensitivity, or whether the nonlinear/conditional measures recover protein-confirmed markers that Spearman misses.

Multi-omics integration enters through the cell-type labels themselves: these are derived from the joint RNA+protein representation via Weighted Nearest Neighbors (WNN, [2]), so the prediction targets reflect both modalities even though the four marker-scoring methods see RNA alone. The protein modality therefore informs both the prediction target and the protein-derived ground truth D_c, introducing a shared dependence. This dependence is identical across all four methods, so, since our claims are comparative, it does not bias the ranking of methods; absolute recovery scores are interpreted only relative to one another, not as standalone accuracies.

Advanced techniques not covered in lectures: Integrated Gradients for neural-network attribution; WNN for multi-modal cell representation. We additionally include a stability analysis motivated by [5].

1.5 What diseases/organisms/tissues will you study?

Healthy human peripheral blood mononuclear cells (PBMCs). PBMCs are the standard benchmarking system for single-cell methods because (i) the cell types are well-defined immunologically, (ii) the major lineages are separated by well-characterized surface proteins, and (iii) the CITE-seq panel measures many of these directly, providing the cross-modal signal our protein-derived ground truth depends on.

1.6 List related work

- [1]: driver gene identification via fate-probability correlation in single-cell data; the conceptual starting point.
- [2]: source of the CITE-seq dataset and the WNN integration method.
- [6] First comparison of mutual information and correlation for gene regulatory networks.

- [7] Building on the work of Steuer et al. delivers a higher accuracy empiric comparison of correlation and mutual information.
- [4]: Integrated Gradients, the attribution method used for our differentiable classifier.
- [5]: cautionary results on saliency-map stability.
- [8]: comparison of statistical tests for driver gene recovery.
- [3]: KSG mutual information estimator.

1.7 How does your project differ from related work?

Existing single-cell DE benchmarks [8] compare statistical tests against each other, all within the linear/marginal regime. The correlation-versus-mutual-information comparison itself has a long history in gene regulatory network inference [6, 7], but that work targets gene–gene dependency for network reconstruction, not the recovery of cell-type markers, and predates modern attribution-based approaches. Our contribution is the factorial decomposition: by including all four cells of the linear/nonlinear $\times$ marginal/conditional grid on the same data with the same ground truth, we can attribute performance differences to specific statistical properties rather than to method-implementation idiosyncrasies. The CITE-seq cross-modal protein validation is, to our knowledge, not used as ground truth in any prior marker-identification benchmark.

2 Data

2.1 Where will you get your data from?

The Hao et al. (2021) PBMC CITE-seq reference dataset [2], available via the NYGC/Seurat reference portal (https://atlas.fredhutch.org/nygc/multimodal-pbmc/) and as a preprocessed AnnData object via scvi-tools tutorials. Cell-type annotations at three levels of granularity are included.

2.2 How many samples are available?

The dataset contains approximately 161,000 cells from 8 donors, with ∼20,000 genes (RNA) and ∼228 surface proteins (ADT) per cell. We will subsample to ∼20,000 to 30,000 cells stratified by cell type to keep MLP training and mutual information estimation tractable on standard hardware.

2.3 Is the data multi-modal (multi-omics)? Is all data coming from the same study?

Yes, two modalities (RNA + ADT surface protein) measured on the same cells in the same study [2]. Samples are paired by construction (shared cell barcode), making integration trivial.

2.4 In what format is the data available?

The data is available as `.h5seurat` and `.h5ad` (AnnData) files. We will work with the AnnData/muon format in Python/R.

2.5 Do you expect to spend a significant amount of time on pre-processing the data?

No. The data is fully preprocessed by the original authors (QC-filtered, normalized, annotated). Estimated < 1 day for data acquisition and sanity checks. Minor additional steps will include rank transformation prior to MI estimation and stratified subsampling, both straightforward.

2.6 Are you sure there are no restrictions on data access?

No restrictions. The data is openly distributed under the Seurat/NYGC reference portal and is freely downloadable without registration.

3 Project Plan

3.1 Summary

Identifying marker genes for cell types is a foundational single-cell analysis task, almost universally performed with one-vs-rest correlation or Wilcoxon tests. These default methods assume marginal, monotonic relationships between gene expression and cell identity. Mutual information relaxes the linearity assumption, partial correlation relaxes the marginality assumption, and gradient-based attribution on a neural classifier possibly relaxes both; but their relative benefit on single-cell data, and which assumption matters more, has not been systematically evaluated.

We will benchmark four dependency measures on PBMC CITE-seq data [2] ($\sim$25k subsampled cells, 8 donors, RNA + $\sim$228 surface proteins): Spearman correlation (linear, marginal), partial correlation (linear, conditional), KSG mutual information [3] on rank-transformed expression (nonlinear, marginal), and Integrated Gradients [4] on an MLP cell-type classifier (nonlinear, conditional). This 2×2 design lets us attribute performance differences to nonlinearity, multivariate context, or their interaction, a question existing DE benchmarks [8] cannot answer because they vary only the test statistic, not the underlying assumptions.

Ground truth is protein-derived: for each cell type we take the surface proteins elevated in that type (one-vs-rest on the ADT modality), map them to their encoding genes, and ask how well each RNA-based method recovers that gene set. The four methods see RNA only; the protein modality builds the ground truth, keeping the test cross-modal. Recovery is scored with the rank-based driver-recovery metric of [1] ($\mathrm{AUC}_{\mathrm{rel}}$), which is parameter-free and defined on the full gene ranking each method produces.

The expected output is not "method X wins" but a characterization of when each statistical assumption matters: per gene, the pattern across the four measures attributes any gap to nonlinearity, confounding, or both, with practical guidance on when the simpler measure suffices and whether the classifier is warranted.

3.2 Timeline

- Pitches take place on 8.6.2026 and project presentation on 13.7/15.7. This leaves 5 weeks for work.
- Code deadline: 24.7
- Report deadline: 31.7
- The week before the pitches is denoted as week 0.
- The time after the presentation is unified to one block.

Week 0 (1.6 - 7.6) - Before pitch

- Pitch preparation; download CITE-seq data; verify cell counts, donor metadata, and annotation levels; set up environment.

Week 1 (8.6 - 14.6) - Pitch

- Project pitch.
- Exploratory analysis: UMAP, QC plots, validation of cell-type labels against RNA and ADT markers.
- Stratified subsampling strategy fixed.

Week 2 (15.6 - 21.6)

- Implement and validate the Spearman correlation and partial correlation baselines (one-vs-rest, per cell type).
- Implement the KSG mutual information pipeline on rank-transformed expression.
- Construct the protein-derived ground truth (elevated surface proteins per cell type, mapped to encoding genes).
- First cross-method comparison on one cell type as a sanity check.

Week 3 (22.6 - 28.6)

- Train the MLP cell-type classifier; hyperparameter sweep; cross-validated accuracy reporting.
- Compute Integrated Gradients attributions, per gene per class.
- Implement the evaluation pipeline against the protein-derived ground truth.

Week 4 (29.6 - 5.7)

- Full benchmark: all four methods across all cell types, scored with the rank-based recovery metric.
- Per-gene decomposition analysis (attribute gaps to nonlinearity, confounding, or both).

Week 5 (6.7 - 12.7)

- Stability and robustness analysis: subsampling, seed variation, sensitivity to MLP architecture.
- Visualizations and interpretation of method-specific genes.
- Prepare presentation.

Week 6 (13.7 - 19.7) - Presentations

- Project presentation (13.7 or 15.7).
- Begin report; integrate presentation feedback.

After presentation (16.7 - 31.7)

- Finalize code repository (README, environment file, reproducibility notebook); code deadline 24.7.
- Write report; integrate optional/stretch goals if time allows (second dataset; alternative attribution method).
- Report deadline 31.7.

Disclosure

Usage of AI. During the preparation of this project pitch, the authors used Anthropic's Claude Opus 4.8 to assist with refining the project scope, pressure-testing the experimental design (in particular the choice of ground truth and the avoidance of data leakage), and drafting and editing the LaTeX manuscript. All scientific decisions, the final framing of the research question, and the validity of the methodology were reviewed and verified by the authors, who take full responsibility for the content.

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